Introduction to Computer System Security

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1 Internet and Networking

The mantra of networking is "anytime , anywhere , anything"

Internet network forms a massive distributed environment.

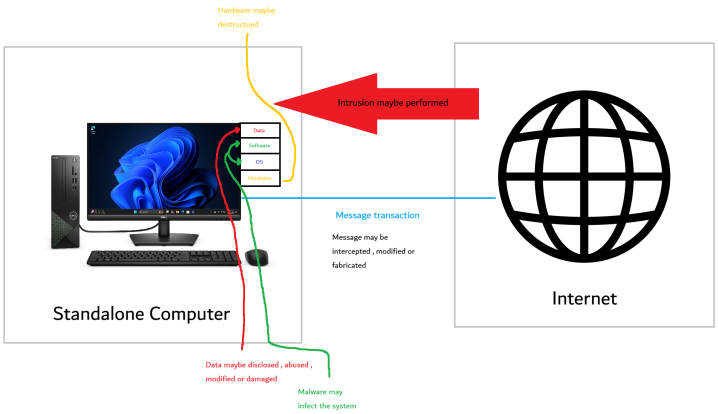
Networking makes system more attractive with attackers.

2 Facets of the Security Problem

In the past, no business is conducted across the Internet. Today, people perform online transactions every day.

More recently, skillful intruders are attacking computers with criminal or military goals in mind.

Encryption is a very important part of network security.



3 Holistic Scene of Security Problem

4 Potential Vulnerabilities

4.1 Categories of Security weaknesses

1. Physical
2. Technology
3. Policy
4. Configuration
5. Human

4.2 Physical

Buildings and equipment rooms may be vulnerable

Intruders can break into server room , and sabotage it

Guards , Locks , Biometric devices must provides a protection against breaks in

4.3 Technology

Every technology has known or unknown inherent weaknesses or vulnerabilities.

Some of this issues is :

- Internet protocols were not designed for security
- Computers and networks operating systems have vulnerabilities
- Network equipments (Routers - Switches - and more...) may be have vulnerabilities must be recognized and protected against

4.4 Policy

Security police weakness is catchall phrase of company policies or lack of polices
Some of issues that can negatively impact the policy :

- No written security policy
- Lack of disaster recover plan
- Lack of security monitoring
- No Policy for hardware and software additions or changes
- Internal policies
- Employment policies

4.5 Configuration

Many network devices have default settings that ease of installation without regard for security issues.

Network administrators need reconfigure the computing devices.

4.5.1 Some common configuration issues include

- User IDs and Passwords in clear text
- Old , missing , duplicate passwords
- Unauthorized or weak remote access through the internet.

4.6 Human

The people who administer and use the computer system represent the greatest vulnerability of all.

Human stupidity, carelessness, laziness, greed, and anger represent the greatest threats to computer system security.